



DRIVECARE PRO

Automotive Mobile App for Vehicle Owner Self-Service

FINAL EXPO / PROJECT PRESENTATION

Database Systems Engineering & Distributed Backend Development
Course: 25CS1302E

P. Bhavan — 2520090146
M. Shiva — 2520090135
N. Sainathreddy — 2520090115

Guide: Dr. Khaja Shareef

KLH Bachupally Campus | AY 2026–2027

Problem Statement & Objectives

2

Why DriveCare PRO was developed

THE PROBLEM

Vehicle owners need one place to manage vehicles, service bookings, service history, roadside help and service-center information.

Scattered maintenance information

Limited booking/status visibility

Difficulty maintaining digital service history

Need for faster roadside assistance

Need for transparent service information

OBJECTIVES

Build a self-service platform backed by a structured MongoDB database and REST APIs.

Secure registration and login

Vehicle registration and health information

Service booking and tracking

Digital service records and bills

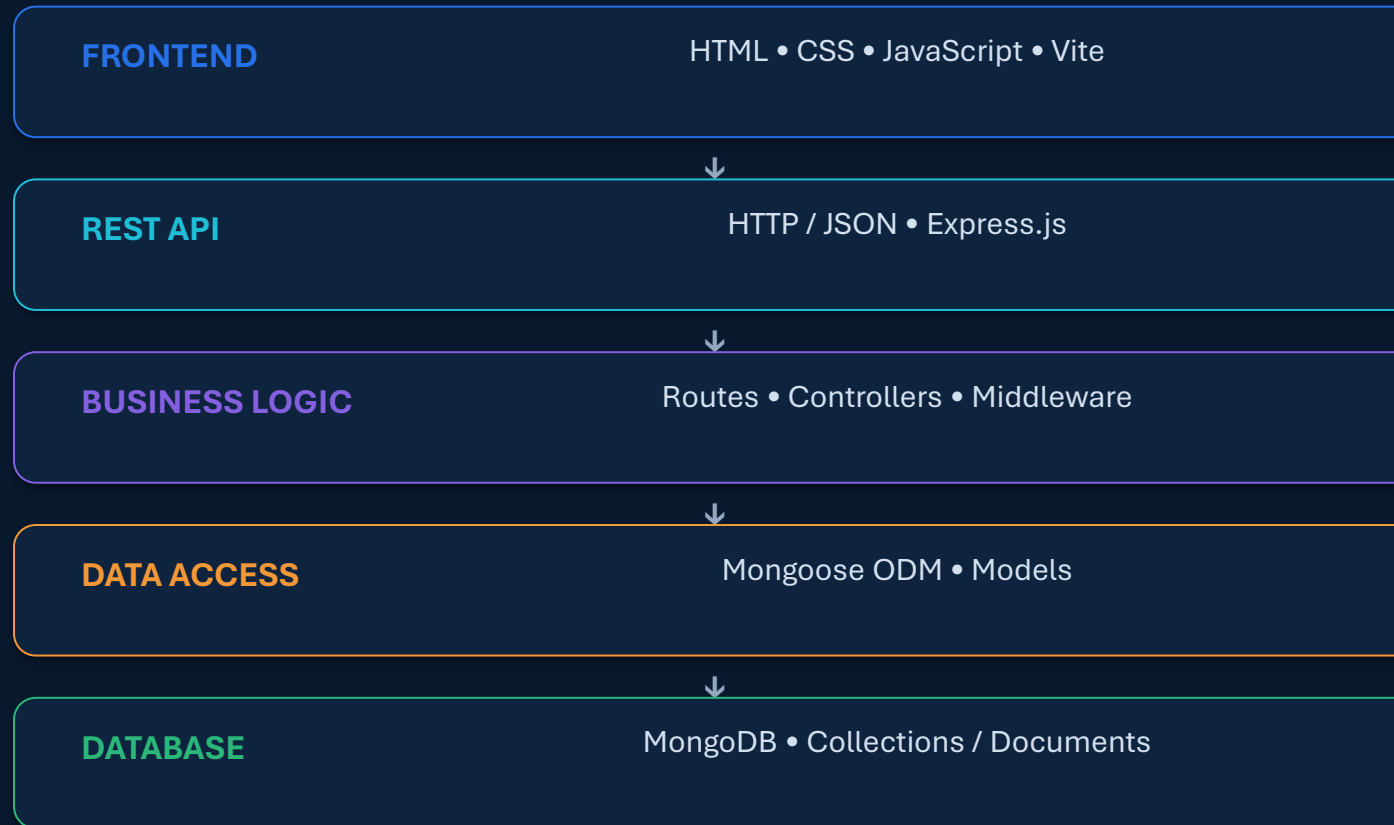
Roadside SOS request management

Notifications and service-center information

System Architecture

3

End-to-end request and database flow



User action → API request → middleware → controller → Mongoose → MongoDB → JSON response

Technology Stack

4

Tools used across the application

Frontend

HTML, CSS, JavaScript, Vite

Backend

Node.js, Express.js

Database

MongoDB

ODM

Mongoose

Authentication

JWT + bcryptjs

Testing

Postman

Deployment

Vercel

Development

Git / GitHub / VS Code

MongoDB Database Design

5

Main collections and relationships

User

Identity, role, contact, profile

Vehicle

Owner reference, details, health

Appointment

Booking, date, slot, status

ServiceRecord

Work, parts, labor, invoice

ServiceCenter

Location, services, bays, rating

RoadsideRequest

SOS type, location, mechanic, ETA

Notification

Updates, type, read state

RELATIONSHIPS

ObjectId references connect users, vehicles, appointments, service records and service centers.

Code & REST API Flow

6

What happens when the user performs an action



CRUD EXAMPLE

Create: register vehicle
Read: view vehicles
Update: change details
Delete: remove vehicle

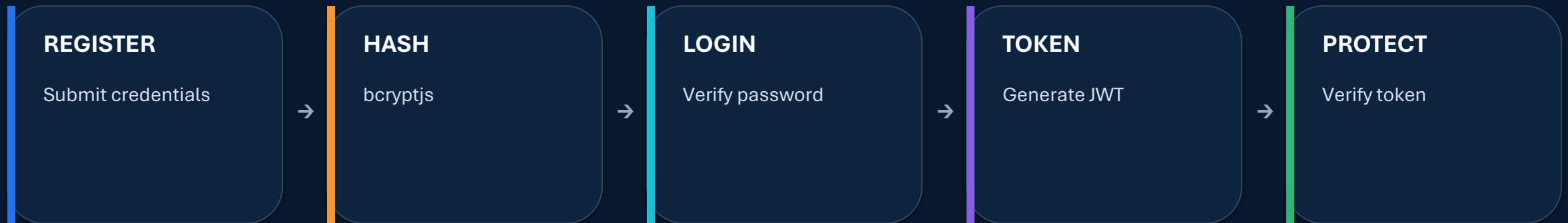
IMPORTANT CONCEPTS

async/await • try/catch • middleware • ObjectId •
ref/populate • HTTP status codes

Authentication & Security

7

Login flow used in the backend



VIVA POINT

Authentication = who you are. Authorization = what you are allowed to do.
Passwords are not stored as plain text; protected API access uses JWT after login.

Core Features

8

Major modules visible in DriveCare PRO

Vehicle Management

Register vehicles and view vehicle health.

Service Booking

Choose service, date/time and track status.

Digital Service Ledger

Store work, parts, labor, tax and invoice data.

Roadside SOS

Create emergency requests with location.

Service Centers

View hubs, services, location and availability.

Notifications

Booking, roadside and service-reminder updates.

Expo Demo Flow

9

Simple sequence to demonstrate the working system

01 LOGIN

Sign in / register

02 VEHICLE

Register or view vehicle

03 BOOK

Create service appointment

04 SERVICE

Show digital service record

05 SOS

Show roadside request

06 DB

Explain MongoDB flow

Remember: Login → JWT → Vehicle → Appointment → Service Record → Notification

Most Likely Technical Viva Questions

10

Keep answers short and confident

Why MongoDB?

Document-oriented and flexible for service/vehicle records.

Why Mongoose?

Schemas, models, validation and easier MongoDB operations.

What is REST API?

HTTP interface using GET, POST, PUT and DELETE.

Why JWT?

Authenticates protected API requests after login.

Why bcrypt?

Hashes passwords instead of storing plain text.

What is middleware?

Runs between the request and the route handler.

What is ObjectId/ref?

References documents in another collection.

What is CRUD?

Create, Read, Update and Delete.

Conclusion & Future Scope

11

What the project demonstrates

CONCLUSION

DriveCare PRO brings vehicle self-service workflows into one application backed by a Node.js/Express REST API and MongoDB database.

Centralized vehicle/service information

Database-backed appointments and records

Secure authentication flow

Roadside assistance management

Service-center and notification support

FUTURE SCOPE

Possible extensions for a production-ready automotive platform.

Real-time vehicle telemetry

Predictive maintenance

Push notifications

Online payments and invoices

Native Android application

Analytics and recommendations



THANK YOU

Questions & Discussion

DriveCare PRO – Automotive Mobile App for Vehicle Owner Self-Service

Project Website: <https://dbms-2nd.vercel.app/>

P. Bhavan • M. Shiva • N. Sainathreddy